

Standing on the 10–20 System: The Right Density for the Right Question

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Additional Declarations: Competing interest reported. Several methods discussed in this review — including sLORETA, eLORETA, the unmixing procedure, and VARETA — were developed by researchers with whom some of the authors have professional or institutional relationships. Specifically, the

unmixing theorem of Biscay, Bosch-Bayard & Pascual-Marqui (2018) involves a co-author of this review. The clinical evidence reviewed in Sections 5.4 and 5.5 was produced by groups with which the authors have collaborative ties. These relationships constitute a potential bias toward positive findings from the methods and pipelines described, and readers should weigh the evidence accordingly. The taxonomic framework and the critical assessment of conditions and limitations were designed explicitly to mitigate this risk, and external critical references (Grave de Peralta Menendez & Gonzalez Andino 1998; Grova et al. 2006; Plummer et al. 2008; Hatlestad-Hall et al. 2023) have been included to provide independent perspectives. No financial conflicts of interest exist. The authors have no shareholding or consultancy relationships with EEG manufacturers or commercial entities related to the content of this review.

Standing on the 10–20 System: The Right Density for the Right Question

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Abstract

This review asks an application-specific question: for which source-level inference tasks does the standard 10–20 EEG system (19 channels) have demonstrated utility, and when does higher-density recording provide a necessary gain? We integrate theoretical analyses, simulation findings, and a structured survey of clinical evidence using a four-level taxonomy defined by spatial scale: scalp/sensor inference (Level 1), lobar source power (Level 2), sub-lobar / predefined parcel-level inference (Level 3; for ROI unmixing specifically, $L < M$ is required for algebraic identifiability), and focal/dipolar localisation (Level 4). The central distinction is between spatial blurring and localisation bias. For sLORETA and eLORETA, the point-spread-function peak is unbiased under the assumed forward model, although the estimate remains spatially broad; with the 10–20 system, localisation bias is not a necessary consequence of low electrode count when these estimators are used; the dominant limitation is spatial resolution (PSF breadth), alongside CTF-mediated contamination from neighbouring generators that is broader at lower electrode densities. Measurement noise narrows the advantage of dense arrays in clinical conditions, while electrode geometry and method choice remain important. Across the clinical literature, the 10–20 system is well supported for sensor-level biomarkers and has demonstrated utility for selected lobar source-power applications under appropriate conditions (signal-to-noise ratio, forward modelling, and reference electrode); equivalence to high-density EEG has not been established for this level. ROI-to-

ROI connectivity at Level 3 is feasible only under explicit dimensionality and conditioning constraints and still requires independent validation. For focal presurgical localisation (Level 4), high-density EEG with individual head models is clearly indicated. Broad ERP components follow the same scale-dependent logic. The principal geometric weakness of the standard 10–20 montage is inferior-surface coverage, which can often be addressed by targeted electrode additions. The key question is therefore not whether more electrodes are better in the abstract, but whether the evidence of demonstrated utility at the required spatial scale justifies the higher-density investment or already warrants it on grounds of patient safety (presurgical localisation).

Keywords: 10–20 system, EEG source estimation, spatial resolution, source visibility, electrode density, quantitative EEG

1. Introduction

Since Herbert H. Jasper codified the international 10–20 system in 1958, the 19 electrodes/channels of this system have remained the de facto standard for clinical EEG (Electroencephalogram) recording (Jasper 1958; Klem et al. 1999). Over seven decades, the system has proved sufficiently robust to classify rhythmic brain activity, detect epileptic abnormalities, describe EEG maturation across the lifespan, and serve as the foundation for cross-cultural quantitative normative databases (John et al. 1977; Harmony et al. 1988; Alvarez et al. 1987). This historical context is important: the 10–20 system should not be evaluated against the requirements of high-resolution voxel-level tomography, for which it was never designed. The design purpose, adoption history, and extensions of the system are reviewed in Section 3. By the same token, high-density EEG should not be dismissed as unnecessarily complex: when sub-lobar or focal precision is the goal, the investment in additional electrodes is well justified. The productive question lies between these poles: for each class of clinical or scientific application, what is the minimum electrode configuration that delivers the required spatial validity, and what does upgrading beyond that threshold actually buy in terms of resolution, reliability, and clinical outcome?

With the emergence of EEG source imaging methods — collectively referred to as inverse solutions or electromagnetic tomography — a widespread view developed that 19 electrodes constitute a technically insufficient configuration for estimating the distribution of cortical source currents. This position is supported by empirical observations that localisation accuracy increases with electrode density (Lantz et al. 2003; Song et al. 2015; Michel & Brunet 2019; Thirairajah et al. 2021), and by theoretical considerations about the Nyquist sampling theorem applied to the scalp electric field. That evidence should not be dismissed: under conditions requiring sub-lobar spatial precision, higher electrode counts provide genuine benefits.

At the same time, the density debate has conflated several phenomena that deserve separate treatment. Reduced spatial resolution (blurring) is not the same as systematic localisation bias: some inverse methods achieve zero localisation bias regardless of sensor count, albeit at the cost of broader spatial spread. The appropriate level of electrode density also depends critically on the target application — spectral biomarker extraction, ROI (Regions of Interest)-level connectivity, and presurgical focal localisation place very different demands on the spatial

resolution of the inverse solution. Furthermore, in typical clinical recording environments, measurement noise increasingly limits the marginal benefit of very dense arrays (Ryynänen et al. 2004). These distinctions have important consequences for how 19-channel systems should be evaluated.

The present work critically reviews the available evidence, aiming to delineate precisely the conditions under which the 10–20 system is sufficient for EEG source estimation, those in which it requires supplementation, and those in which higher-density recording is genuinely indicated. Our central argument is not that the 10–20 system is universally adequate — it is not — but that its applicable range is broader than its reputation in high-resolution source imaging contexts might suggest. The most consistently documented limitation is incomplete inferior surface coverage; for focal Level-4 localisation, both geometric and sampling constraints apply.

2. Review Methods and Definitional Framework

A complete glossary of abbreviations and technical terms is provided in Supplementary Section SG. Extended mathematical definitions, derivations, and method-family details are provided in Supplementary Sections S1–S5.

2.1. Search strategy and inclusion criteria

This review employed a structured multi-database search following PRISMA 2020 reporting principles (Page et al. 2021), adapted for a critical narrative review with pre-specified inclusion and exclusion criteria.

Four databases were searched on 6 August 2026 (PubMed, Scopus, IEEE Xplore, and Google Scholar), covering 1980–August 2026. Six thematic blocks addressed electrode configuration, spatial resolution and leakage, source connectivity and resting-state networks, epilepsy, normative quantitative EEG (qEEG) biomarkers, and evoked responses.

The search retrieved 6,573 records; 3,199 unique records remained after DOI/title deduplication. Full-text assessment of 152 records yielded 117 eligible studies incorporated in the evidence synthesis. A further 14 foundational, methodological, or historical references cited in the paper (including pre-1980 works, the PRISMA 2020 guidance, and a directly relevant book chapter on eLORETA density effects) were added outside the search scope; these are identified in Supplementary Table S0.4. The screening flow is shown in Figure 1. For references carrying central conclusions of the review, we additionally assessed claim-to-source concordance: each reference was classified as directly supporting the stated claim, partially supporting it under narrower conditions, or not supporting the claim as worded. References providing contrary evidence were retained explicitly to define the boundaries of inference (Supplementary Table S6c). Detailed search blocks, inclusion/exclusion criteria, and the complete reference list are provided in Supplementary Section S0 and Table S0.4; study-level evidence is tabulated in Supplementary Section S6.

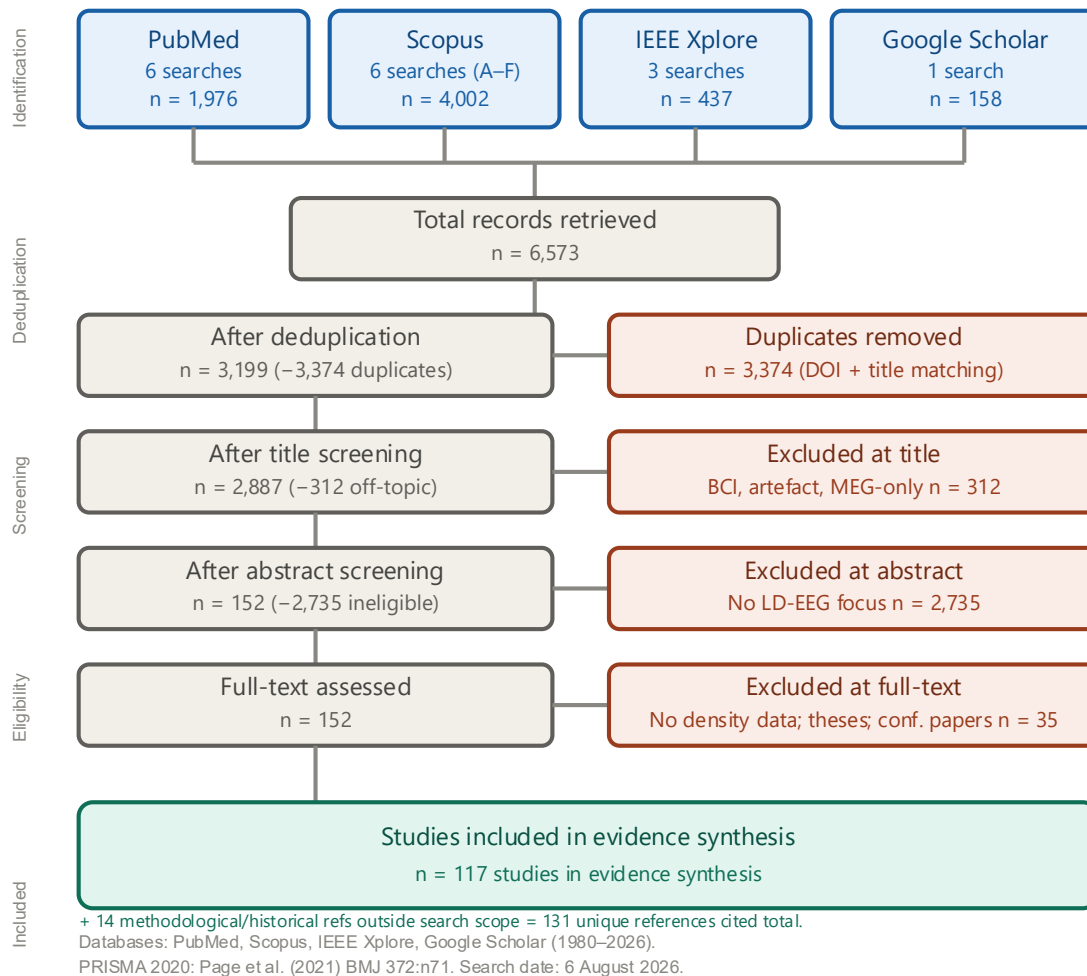


Figure 1. PRISMA 2020 flow diagram. Records retrieved from four databases (1980–2026); 6,573 total after search. After deduplication (n=3,199), title and abstract screening (n=152 for full-text assessment), 117 studies were included in the evidence synthesis. An additional 14 foundational/methodological references (pre-1980 works, the PRISMA 2020 guide, and a book chapter on eLORETA density effects) were added outside the search scope; all are identified in Supplementary Table S0.4. Total unique references cited: 131.

Detailed mathematical derivations are intentionally kept out of the main text because the primary aim is conceptual and clinical synthesis. Supplementary Sections S1–S5 provide the common mathematical framework and expanded method-family descriptions; primary derivations are cited throughout.

2.2. Taxonomy of validity levels

A fundamental conceptual problem in the electrode-density debate is that 'validity for EEG source estimation' is not a unitary concept. We organise the taxonomy around a single primary axis — spatial scale — ranging from sensor-level scalp inference to focal sub-lobar source localisation. Within each spatial level, signal types (resting-state spectra, evoked potentials, interictal spikes, connectivity measures), noise sources, and clinical applications are discussed as cross-cutting dimensions in Section 5, rather than being embedded in the level definitions. This yields four levels of inference with different spatial resolution requirements, different gold standards, and different sensitivities to electrode count. To avoid conflating them, we defined a validity taxonomy (see Table 1) that we use throughout this review.

This taxonomy explains why the same electrode montage can be valid at one level and insufficient at another. The four levels are defined purely by spatial resolution target: scalp/sensor (Level 1), lobar (Level 2), sub-lobar / predefined parcel-level (Level 3), and focal/dipolar (Level 4). Within each level, the relevant signal types (resting-state spectra, evoked potentials, connectivity measures, interictal spikes), SNR (Signal to Noise Ratio) requirements, noise sources, and clinical applications are discussed as cross-cutting dimensions in Section 5 — rather than being used to define the levels. In particular, connectivity (iCoh, PLV, DTF, PDC) is treated as a measure type applicable within Level 3, not as a level in itself. The conclusions of this review apply to specific levels of taxonomy, not to EEG source estimation in general. When we argue that 19 electrodes can ‘support’ a given level of inference, we mean that peer-reviewed evidence demonstrates fit-for-purpose performance at Levels 1–2 under appropriate conditions. This does not entail that more electrodes would not improve performance, nor that 19 electrodes are equivalent to higher-density systems at those levels. At Level 3 the evidence is conditional; at Level 4, higher density is clearly indicated.

Table 1. Validity taxonomy for EEG source estimation, organised by spatial-resolution scale. Signal types, clinical applications, and Evidential assessment for S10-20 are assessed within each spatial level as cross-cutting dimensions, rather than defining the levels themselves.

Level	Spatial scale / Resolution target	Signal types and measures	Clinical applications	Evidential assessment for S10-20
Level 1 Scalp / Sensor	Sensor-level; no source inversion required. Spatial: scalp topography or scalp-average spectral measure. Resolution: not applicable (no source model).	Resting-state band power, spectral ratios (TBR, alpha/theta), electrode coherence, global scale indices. Broad ERP components (P300, N400, auditory N1) analysable at scalp level.	Normative qEEG databases; age-related maturation indices; dementia and MCI screening biomarkers; neurofeedback; batch-effect correction (Global Scale Factor, HarMNqEEG).	19 electrodes can support this level of inference. The main requirements are noise control, batch correction, and age-matched norms. Independent and author-group evidence converge.
Level 2 Lobar	Source-level inference targeting lobar or large predefined cortical regions; no claim of within-lobe localisation. Method-specific spread with appropriate inverses: SD $\approx$ 20–50 mm (eLORETA/sLORETA); FWHM $\approx$ 30–35 mm (cLORETA/VARETA, volumetric metric; see footnotes [a–d] in Table 2). These are properties of specific methods, not definitional criteria for Level 2.	Lobar source power spectra (delta, theta, alpha, beta); source-level ERPs with broad scalp distributions (P300, N1, ASSR); test-retest reliability of source estimates; band-specific spectral biomarkers at source level.	Alzheimer’s disease and MCI biomarkers (hippocampal–alpha correlations); epilepsy lobar concordance (presurgical screening); developmental EEG maturation norms; qEEG tomographic norms (CHBMP, HarMNqEEG).	S10-20 can support lobar-scale inference under appropriate inverse methods and head models; equivalence to high-density EEG has not been established. Sub-lobar anatomical precision is not attained. Key assumptions: adequate SNR, a valid forward model, and appropriate reference.
Level 3 Sub-lobar / predefined parcel- level	Source-level inference among predefined sub-lobar parcels or regions. Intended spatial scale: parcel size ($\sim$ 2–4 cm diameter). Leakage and cross-talk must be explicitly considered. These spatial criteria define Level 3 independently of any particular inverse or unmixing method.	ROI-to-ROI effective and functional connectivity (iCoh, PLV, DTF, GC); source-level directed connectivity networks; spatially focal ERP components (N170, MMN in restricted populations) at the boundary of this level.	Paediatric neurodevelopmental disorders (dyslexia, ADHD): directed connectivity hubs. Normative connectome development (infancy, ageing). Psychiatric conditions where network-level biomarkers are sought.	Conditional evidence. For ROI unmixing (Biscay et al. 2018), algebraic identifiability requires $L < M$; with S10-20 this permits up to $L \leq 18$ scalar ROI time series for known/fixed source orientation, or $L \leq 6$ for unconstrained 3-D vectors. This is a method-specific identifiability condition, not a definition of Level 3. Independent replication of the unmixing pipeline remains limited; blind network decomposition (ICA/sICA) generally requires higher

				density. Evidence at this level remains primarily author-group and proof-of-concept.
Level 4 Focal / Dipolar	Voxel-level or equivalent; sub-lobar and focal precision (< 2 cm). Full distributed source map or dipole fit. Resolution target: individual gyrus or sulcus.	Interictal and ictal spike localisation; early focal ERP components (N170, early sensory components); focal oscillatory sources. Note: subcortical source detection depends primarily on source geometry (open- vs closed-field), depth, and visibility — not on spatial scale per se — and is not a defining feature of Level 4.	Presurgical epilepsy planning (concordance with intracranial EEG, resection boundaries); focal cortical dysplasia localisation; precise motor or language mapping for surgical safety.	S10-20 are insufficient for routine focal sub-lobar localisation. 64–256 electrodes with individual MRI are clearly indicated. 19-electrode systems may support lobar-level concordance but not focal (< 2 cm) precision.

3. Historical Context: The 10–20 System and Its Extensions

The 10–20 system was the product of a consensus process initiated at the First International EEG Congress in London in 1947, where Herbert H. Jasper was charged with studying methods to standardise electrode placement. Jasper presented the resulting system at the Fourth International EEG Congress in Brussels in 1957 and published it formally in 1958 (Jasper 1958). The International Federation of Electroencephalography and Clinical Neurophysiology adopted it as the international standard that same year — a status it retains to this day.

It is essential to note that the 10–20 system was designed for a specific and concrete purpose: to provide a reproducible electrode placement method guaranteeing that each position covers the same brain regions across subjects, regardless of head size, and that results could be compared across laboratories. It was not designed for high-density sampling of the scalp electric field or for computational tomography, disciplines that did not exist in 1958. That the system also works for these applications is therefore a later, unanticipated capability rather than a design specification.

The 10–20 family has generated higher-density extensions — the 10–10 system (Chatrian et al. 1985; Nuwer et al. 1998) with up to 74 positions, and the 10–5 system (Oostenveld & Praamstra 2001) with up to 345 positions — all of which are enlargements built upon Jasper's original grid rather than replacements. Anatomical studies have confirmed that C electrodes in the 10–20 system are located just 1 cm within the central sulcus (Klem et al. 1999; Koessler et al. 2009), underscoring their consistent neuroanatomical relevance. The updated IFCN guidelines (Seeck et al. 2017) acknowledge the 10–20 system as the clinical standard and propose expanding it to 25 electrodes solely through the addition of the inferior temporal chain — the anatomical limitation of the original system — without mandating a complete replacement. The same guidelines specify, however, that for EEG source localisation the 10–10 extension (or partial 10–10) is recommended, and that 64–128 or more channels are appropriate for high-resolution ESI. This tiered recommendation is precisely consistent with the task-dependent framework proposed here: the 10–20/25-electrode range supports Levels 1–2, while Level 4 presurgical localisation warrants higher-density recording. The role of the 10–20 system as the basis for multinational normative qEEG databases over five decades is reviewed in Section 6.

4. Mathematical Foundations: Spatial Resolution, Localisation Bias, and Unmixing

4.1. Blurring versus localisation bias: a critical distinction [Theoretical result]

The EEG inverse problem — estimating the distribution of cortical source currents from scalp-recorded potentials — is fundamentally ill-posed: it has no unique solution. Any inverse method necessarily introduces additional constraints (regularisation) that determine the resulting solution. A mathematically unavoidable consequence is that for linear inverse estimators, the source estimate at any given voxel is a weighted linear combination of the true sources across the cortex — the so-called mixing or blurring effect, whose spatial extent is

quantified by the width of the point spread function (PSF), often summarized by measures such as full-width at half-maximum (FWHM) or spatial dispersion (Hedrich et al. 2017; Nguyen-Danse et al. 2021).

It is critical to distinguish this blurring from localisation bias, which refers to the displacement of the estimator maximum relative to the true source location (see Figure 2). Pascual-Marqui (2007) demonstrated the existence of a family of linear imaging methods — including sLORETA and eLORETA — with the property of zero localization: the bias of the PSF under the assumed forward model is null in the presence of both measurement noise and structured biological noise. Electrode count primarily affects spatial resolution, that is, the width of the spread function, rather than the formal zero-bias property of these estimators under their model assumptions, which holds mathematically irrespective of the number of sensors.

However, several qualifications are essential to avoid misreading this result clinically. First, zero localisation bias does not imply high spatial precision. With S10-20, sLORETA/eLORETA remain spatially broad: reported spread is approximately SD 20–50 mm (see Table 2; Supplementary S2) — the peak is correctly centred on the true source, but the estimate is spatially diffuse. Note that SD and FWHM are not interchangeable; FWHM $\approx$ 30–35 mm applies to cLORETA/VARETA (volumetric metric; see Table 2, footnote [d]). This is adequate for lobar-level analysis but insufficient for reliable sub-lobar discrimination. Second, the zero-bias property is a statement about the PSF, that is, the reconstruction of an isolated source. It is not a statement about the cross-talk function (CTF), which quantifies leakage from other active locations into the estimate at the voxel of interest: each source estimate also contains weighted contributions from other active sources, and this cross-contamination is generally not zero even for eLORETA/sLORETA-type estimators (Grave de Peralta Menendez & Gonzalez Andino 1998). Third, the zero-bias result depends on the accuracy of the forward model: errors in the head model, electrode co-registration, or conductivity assumptions introduce effective localisation bias that is not corrected by the inverse method. Fourth, 'zero error localisation' in the mathematical sense refers to the expected peak location under the model assumptions; it does not constitute a guarantee of anatomical accuracy in any individual recording.

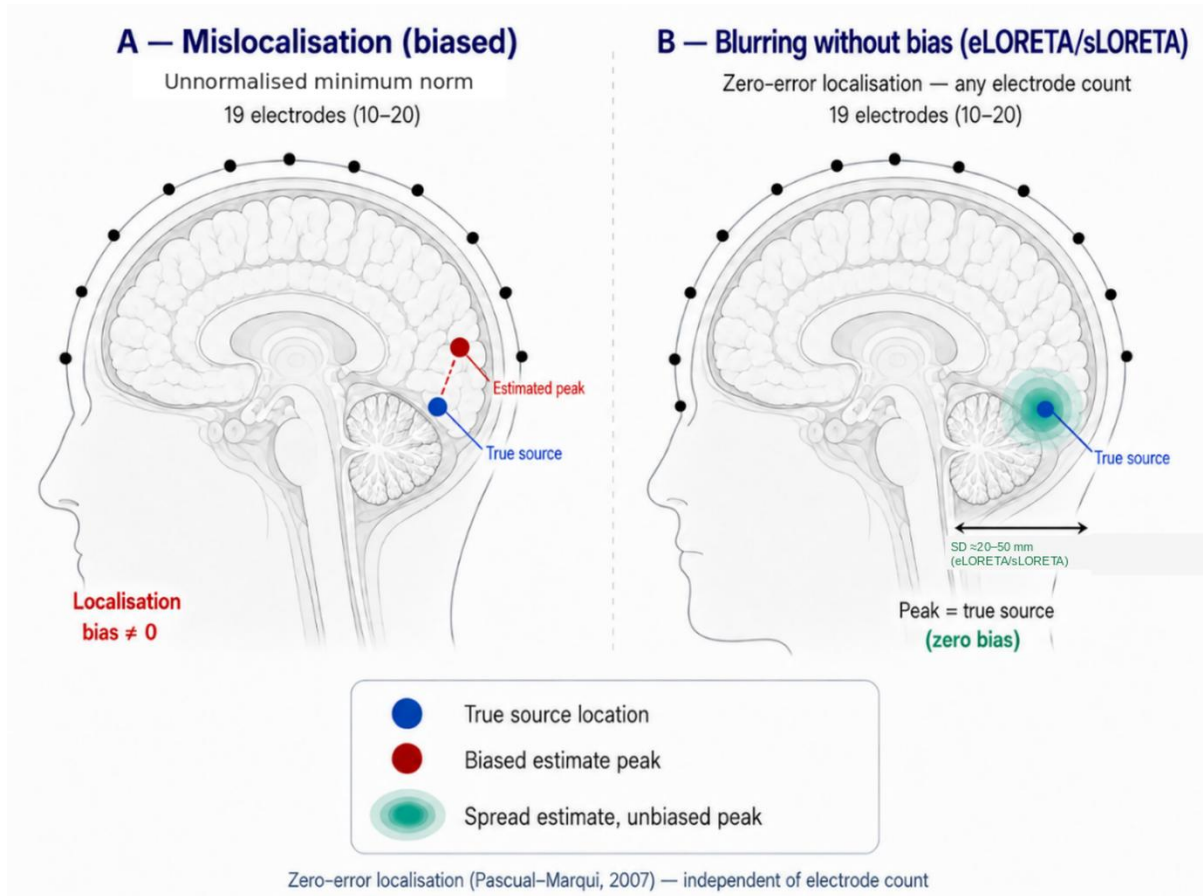


Figure 2: Blurring versus localisation bias in EEG inverse solutions. Panel A: a biased inverse solution (e.g. unnormalised minimum-norm estimate with a sparse array) displaces the estimated peak away from the true source — a localisation error that cannot be reduced by signal averaging or increased SNR. Panel B: a zero-error localisation method (eLORETA, sLORETA) produces a spatially spread estimate whose peak coincides with the true source at the cost of substantial spatial blurring ($SD \approx 20\text{--}50$ mm for eLORETA/sLORETA; $FWHM \approx 30\text{--}35$ mm for cLORETA/VARETA (volumetric metric; see Table 2 footnote [d]); metrics are not interchangeable). This spread corresponds to lobar-regional resolution, consistent with Level 2 (lobar source-power) applications, but insufficient for sub-lobar or focal source analysis (Levels 3–4). Level 1 (sensor/scalp inference) does not require source inversion and is therefore not constrained by PSF breadth.

With those caveats established, the distinction between blurring and bias remains pedagogically important. Criticisms of the 10–20 system that state 'localisation is poor with 19 electrodes' often implicitly assume biased inverse methods, such as unnormalised minimum-norm estimate, for which the criticism is valid. For eLORETA and sLORETA, low sensor count does not necessarily induce systematic PSF peak displacement in these estimators: the zero-DLE property holds irrespective of electrode count under the stated model assumptions. Whether the resulting spatial spread remains adequate for a particular lobar application is an empirical question that depends on source configuration, forward-model accuracy, and noise level — and is addressed by the simulation and clinical evidence reviewed in Sections 4.2 and 5.

An important qualification applies throughout: the PSF/CTF metrics and blurring estimates in this section characterise classical regularised linear estimators and should be understood as upper bounds. Methods incorporating strong biophysical or structural priors — such as ξ - α NET (García Reyes et al. 2026) — can achieve substantially narrower effective resolution

from 19-electrode data, as illustrated in Supplementary Section S7.2; although this very recent result has not been replicated by an independent group yet.

4.2. Quantifying the blurring: what price is paid with 19 electrodes? [Simulation evidence]

Simulation studies have quantified the blurring and localisation error associated with different electrode densities across several linear distributed inverse methods. Table 2 summarises results for the LORETA family (sLORETA, eLORETA, cLORETA/VARETA, LORETA), which are the methods on which the normative qEEG literature reviewed in this paper is built, plus MNE and dSPM from Hedrich et al. (2017) for comparison. The focus on the LORETA family is not arbitrary: these methods were selected for normative work precisely because sLORETA and eLORETA achieve zero DLE by construction — a property not shared by MNE or dSPM — making them more appropriate for detecting lobar-level spectral differences in Levels 1–2 applications.

Table 2. Spatial resolution of linear inverse solution methods as a function of electrode/channel density

Method	Modality / Metric (PSF)	19 channels (ldEEG)	120–256 channels (hdEEG)	Reference
sLORETA	EEG / DLE = 0 (exact, by construction); SD ~35–50 mm	DLE = 0; SD ~45 mm [a]	DLE = 0; SD ~35 mm [b]	Nguyen-Danse et al. (2021) [a]; Hedrich et al. (2017) [b]
cLORETA / VARETA	EEG / equivalent radius of volumetric FWHM of the PSF (simulation)	~30–35 mm equivalent radius [d]	~20–32 mm equivalent radius [d]	Trujillo-Barreto et al. (2004) [d]
LORETA	EEG / DLE (non-zero)	Not directly reported [g]	—	See note [g]
eLORETA	EEG / DLE = 0 (under model assumptions; not anatomical accuracy); SD ~20 mm in an 18-channel macaque scalp-EEG condition	DLE = 0; SD ~20 mm [c]	—	Todaro et al. (2019) [c]
MNE	EEG / DLE > 0 (depth bias); hdEEG SD reported by Hedrich et al. (2017)	Not directly reported [e]	DLE ~0–10 mm; SD ~40 mm	Hedrich et al. (2017) [e]
dSPM	EEG / DLE > 0 (noise-normalised MNE); hdEEG SD reported by Hedrich et al. (2017)	Not directly reported [f]	DLE ~0–8 mm; SD ~40 mm	Hedrich et al. (2017) [f]

All values are for EEG under idealised single-source conditions unless otherwise noted; performance with multiple simultaneous sources, inaccurate head models, or high noise levels will differ. hdEEG: high-density EEG (≥ 120 channels). —: data not available for this configuration. [a] Nguyen-Danse et al. (2021) compared 19- vs 128-channel EEG for source-functional connectivity; sLORETA SD ~45 mm is derived from their 19-channel condition. Their primary finding was that a beamformer outperformed sLORETA at 19 channels for recovering behavioural associations, so this value indicates conditional feasibility rather than general adequacy of sLORETA at 19 channels. [b] Hedrich et al. (2017) used 256-channel EEG

and MEG; the sLORETA value shown in the hdEEG column is appropriate to that condition only. [c] Todaro et al. (2019) used 18 scalp EEG electrodes in a macaque simultaneous ECoG/EEG model, not standard human clinical EEG; the eLORETA SD ~ 20 mm is extracted from that specific condition and should be interpreted accordingly. [d] Trujillo-Barreto et al. (2004) report FWHM as a volume of PSF voxels above half-maximum, normalised to sphere volume. The millimetre values shown here are the equivalent-sphere radii derived from that volumetric FWHM; they are not linear FWHM widths and should not be compared directly with SD. [e] Hedrich et al. (2017) studied 256-channel EEG and MEG and did not include a 19-channel MNE condition; therefore no low-density numerical value is reported here. [f] The same limitation applies to dSPM: Hedrich et al. (2017) provide hdEEG values but no 19-channel condition. [g] Vega-Hernández et al. (2008) is a mathematical methods paper on penalised least squares and does not provide a systematic 19-channel LORETA DLE estimate; no low-density numerical value is therefore reported in this table.

The spatial spread metrics in Table 2 indicate lobar-regional resolution with S10-20: SD ≈ 20 –50 mm (eLORETA–sLORETA) and FWHM ≈ 30 –35 mm (cLORETA/VARETA; volumetric metric, see footnote [d]); the provenance of each estimate is detailed in the table footnotes [a–d]. Note that SD and FWHM are not interchangeable (see Supplementary S2); both indicate resolution adequate for normalised qEEG spectral analysis (Level 1) and population-level lobar-source comparisons, but insufficient for presurgical focal localisation (Level 4). Simulation and experimental studies show that eLORETA localisation accuracy improves monotonically with electrode count (18→64→173 channels; Dattola et al. 2020), and that the greatest practical gain occurs when increasing from ~ 30 to ~ 64 electrodes, with diminishing returns above 64 (Lantz et al. 2003; Sohrabpour et al. 2015), illustrating a law of diminishing returns. These findings should be interpreted in the context of our taxonomy: S10-20 may achieve adequate resolution for the intended application even if they fall short of the theoretical optimum. An important limitation that remains unquantified for the lobar case is cross-talk function (CTF) contamination of power estimates: although DLE = 0 ensures the PSF peak is correctly centred, each source estimate integrates weighted contributions from neighbouring generators. At S10-20 the CTF is broader than at higher densities, meaning that an estimate labelled “frontal lobe power” may contain appreciable contributions from adjacent parietal or temporal sources. The studies available (Dattola et al. 2020; Nguyen-Danse et al. 2021) demonstrate that localisation accuracy improves with density but do not directly quantify CTF-mediated power leakage at the lobar scale under typical clinical noise levels. Matched 19-vs-64 comparisons of lobar source-power estimates in physiologically validated cohorts would close this gap and constitute the most informative experiment for the Level 2 claim.

The accuracy of ESI at any electrode density is further modulated by the choice of head model. The choice of head model is well established to affect ESI accuracy: patient-specific models outperform spherical and template approaches, particularly for focal and deep sources; the magnitude of this benefit depends on anatomy, source depth, modelling assumptions, and electrode configuration (Valdés-Hernández et al. 2009; Darvas et al. 2004). For neonates and infants, the challenges of neonatal head anatomy — thin skull, immature tissue conductivities, and complex fontanelle geometry — require age-specific volume conductor models;

individualized ESI pipelines for this age group have been described in the paediatric source imaging literature.

Three independent benchmarking studies provide critical external context for the resolution metrics in Table 2. Hauk et al. (2011) compared noise-normalised minimum norm estimates in MEG using multiple resolution metrics simultaneously — PSF, CTF, localisation error, and spatial dispersion — demonstrating that no single metric captures the full picture of inverse solution performance and that method comparison results are sensitive to which metric is prioritised. Their framework directly informs how Table 2 should be read: SD and DLE provide complementary information, and a method with $DLE = 0$ (sLORETA, eLORETA) can still have high SD, which is the practically relevant limitation for distinguishing nearby sources. Grova et al. (2006) evaluated EEG localisation methods using realistic simulations of interictal epileptic spikes, incorporating physiological noise and realistic head geometry across five inverse methods. Their findings show that the relative ranking of methods changes substantially depending on noise level and source depth, and that performance metrics obtained under ideal simulation conditions may not transfer to clinical recordings — a caveat that applies equally to the low-density case. Yao and Dewald (2005) provided a systematic empirical comparison of source localisation methods using both simulated dipoles and actual somatosensory and motor evoked potentials recorded with 163 electrodes and subsampled to lower densities, finding that the interaction between method choice and electrode density determines outcome more than density alone — a conclusion directly relevant to the argument that method selection is at least as important as electrode count when working within the 10-20 system.

From a methodological standpoint, Soler et al. (2020) demonstrated neural activity reconstruction from low-density EEG using multivariate empirical mode decomposition (MEMD), showing that adaptive decomposition methods can partially mitigate spatial undersampling. Pizzagalli (2007) provides a comprehensive tutorial on EEG and high-density electrophysiological source localization covering the theoretical basis of density effects, spatial resolution, and LORETA-family methods; Baillet (2017) reviews MEG for brain electrophysiology and clarifies the complementary — rather than redundant — relationship between MEG and EEG source imaging, a distinction relevant to interpreting density-comparison studies that include MEG as a reference.

4.3. A taxonomy of inverse solution methods: structure, domain, and source visibility

Electrode-density effects cannot be interpreted independently of the inverse method. The LORETA family is central to the normative qEEG literature reviewed here, but EEG/MEG inverse solutions span minimum-norm, beamformer, Bayesian, sparse/parametric, component-tomography, ROI-unmixing, and generative families (Knösche & Haueisen 2022). These families differ in target domain, priors, effective dimensionality, source visibility, and therefore sensitivity to sensor count.

For the present argument, three distinctions matter most. First, methods that constrain effective dimensionality—through sparsity, parametric source models, component structure, or predefined ROIs—can be more robust to sparse sampling than unconstrained distributed

inverses. Second, cross-spectral or generative approaches (e.g., bcVARETA, ξ - α NET, DCM) fit aggregated spectral or model parameters rather than instantaneous voxel time series and may therefore depend less strongly on high-spatial-frequency scalp information. Third, source visibility depends jointly on depth, geometry, synchrony, noise, and forward-model accuracy; electrode count is only one component of this interaction (Vega-Hernández et al. 2008).

These distinctions prevent a single density rule from being applied to all inverse methods. Focal Level-4 questions favour high-density arrays and constrained focal methods; broad distributed resting-state activity and lobar qEEG require different priors and different performance criteria. A detailed comparison of method families, including STTONNICA, sparse Bayesian methods, beamformers, ROI unmixing, and generative models, is provided in Supplementary Section S5. High-density EEG also offers advantages that are not captured by localisation-error metrics alone: additional spatial degrees of freedom improve ICA-based separation of neural and artefactual components, help distinguish simultaneously active nearby generators, reduce spatial aliasing in distributed network analyses, and sample focal or sulcal scalp fields more completely. These gains are especially relevant to blind network decomposition and focal Level-4 questions (Michel & Brunet 2019; Hassan & Wendling 2018; Liu et al. 2018).

Method choice can therefore alter the practical value of a given montage. Minimum-norm methods are especially vulnerable to depth bias and broad spatial dispersion; sLORETA/eLORETA remove peak-location bias under the assumed model but retain broad PSFs; beamformers can achieve lower DLE for isolated focal generators but are sensitive to correlated or extended sources; sparse Bayesian and parametric methods can sharpen genuinely focal activity at the cost of priors that are inappropriate for diffuse resting-state oscillations. ROI-unmixing and cross-spectral/generative models reduce the effective dimensionality in different ways. These are not interchangeable solutions, and the electrode-density question must be posed jointly with the source model and target domain.

Independent evaluations reinforce this point. Grave de Peralta Menendez and Gonzalez Andino (1998) showed that apparent precision from minimum-norm-type solutions can be misleading when regularisation and non-uniqueness are ignored. Mahjoory et al. (2017), comparing templates, head models, inverse methods, and software implementations, found source localisation to be more stable than source connectivity, indicating that the application domain itself changes sensitivity to method and model choice. The expanded taxonomy and source-visibility discussion are retained in Supplementary Section S5.

4.4. The measurement noise ceiling: why high-density advantage diminishes in clinical settings [Theoretical/simulation evidence]

Electrode density and measurement noise interact strongly in EEG source reconstruction. Ryyänänen et al. (2004) examined this using singular value decomposition (SVD) of the forward transfer matrix in a three-layer spherical head model, comparing configurations from approximately 21 to 512 electrodes. Their analysis showed that the number of recoverable spatial basis vectors depends jointly on electrode count and relative noise level ($NL = \text{measurement noise amplitude} / \text{EEG signal amplitude}$). In nearly noiseless conditions, spatial resolution is mainly limited by electrode count, and denser arrays provide progressively greater

recoverable spatial information. As measurement noise increases, however, finer spatial components fall below the noise floor, and the advantage of very dense arrays diminishes.

Ryynänen et al. estimated typical EEG measurement conditions at approximately $NL = 0.05$ – 0.125 , based on electrode noise of about 2 – 5 μV rms and EEG amplitudes of roughly 10 – 100 μV . Within this range, their simulations placed the best attainable spatial resolution approximately within the 64 – 128 electrode regime, whereas exploiting 256 – 512 electrodes required substantially lower noise levels.

Table 3. Optimal electrode system as a function of relative noise level (Ryynänen et al. 2004)

<i>Relative noise level</i>	<i>Density providing useful/near-optimal spatial resolution</i>	<i>Interpretation</i>
$NL < 0.02$	512	Very-low-noise conditions; very high density can still be exploited
$0.02 \leq NL < 0.04$	256	Low-noise research conditions
$0.04 \leq NL < 0.10$	128	Range overlapping realistic clinical recordings
$NL \geq 0.10$	64	Higher-noise clinical recordings; returns above 64 diminish strongly
$NL \gg 0.10$	Density advantage becomes small	Extreme-noise regime; no evidence of 10–20 sufficiency

NL: relative noise level = electrode noise / EEG signal amplitude. Typical clinical environment: $NL \approx 0.05$ – 0.125 .

The result should not be interpreted as evidence that 19–21 electrodes are equivalent to denser systems. Ryynänen et al. explicitly attributed the poor spatial resolution of the conventional 10–20 system primarily to its limited spatial sampling and concluded that more electrodes are preferable when accurate source reconstruction is required. Their results therefore support two complementary conclusions: sparse arrays are limited mainly by electrode count, whereas increasingly dense arrays become progressively limited by measurement noise.

This distinction is relevant to the task-dependent framework proposed here. For coarse spatial inferences, such as global or regional spectral abnormalities, hemispheric asymmetries, or lobar-level source estimates (Levels 1–2), the additional fine spatial information provided by high-density systems may not materially change the clinical conclusion. Under these conditions, the noise-ceiling effect strengthens the practical case for conventional clinical arrays. In contrast, for sub-lobar discrimination, focal source localisation, closely spaced generators, or presurgical applications (Levels 3–4), the missing spatial information becomes critical, and the noise-ceiling argument cannot be used to justify 19-electrode sufficiency.

These numerical thresholds should nevertheless be interpreted cautiously. Ryynänen et al. used a simplified spherical head model and considered measurement noise without fully incorporating anatomical modelling error, electrode-position uncertainty, tissue conductivity variability, source depth, or inverse-method dependence. The broader principle is therefore more important than any specific electrode count: low-density arrays are sampling-limited, whereas very-high-density arrays progressively become noise-limited.

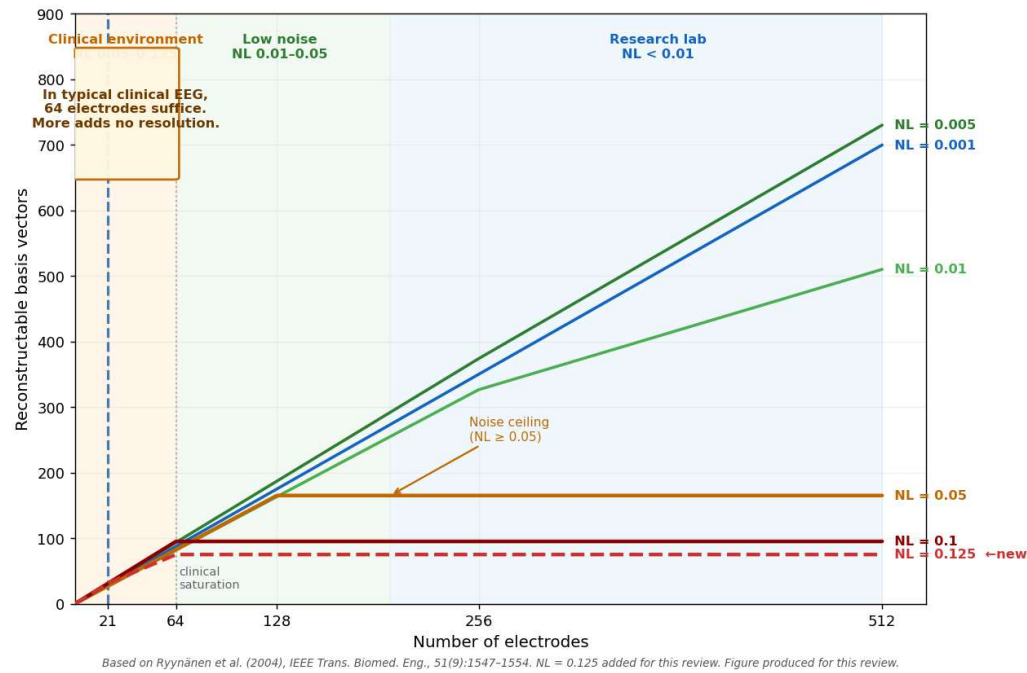


Figure 3. Measurement-noise ceiling and the recoverable spatial information of EEG arrays. Conceptual representation of the interaction between electrode density and relative measurement noise based on Ryyänänen et al. (2004). In nearly noiseless conditions, the number of recoverable spatial basis vectors increases with electrode count. As relative measurement noise rises, fine spatial modes from dense arrays fall below the recoverability threshold, progressively reducing the marginal advantage of 128-, 256-, and 512-electrode systems. Ryyänänen et al. estimated normal EEG measurement conditions at approximately $NL = 0.05$ – 0.125 , a range in which their simulations favoured approximately 64–128 electrodes for maximal recoverable spatial resolution. The convergence of the curves at high noise should not be interpreted as evidence that 19–21 electrodes are spatially equivalent to denser systems.

4.5. Effective dimensionality as a determinant of robustness in low-density EEG [Theoretical/empirical evidence]

A central factor governing performance at low electrode density is the effective dimensionality of the inverse problem. Methods that restrict the number of degrees of freedom through parametric assumptions, sparsity, or structured projections reduce ill-posedness and can be more robust to sensor loss.

Qin, Tong & Sun (2026) tested eight inverse methods under electrode-degradation scenarios using simultaneous sEEG and 256-channel EEG with intracranial stimulation as ground truth. Methods with constrained effective dimensionality (dipole fitting, RAP-MUSIC, MxNE) were substantially more robust to electrode loss than unconstrained weighted minimum norm; the effect was strongest when electrodes near the source were removed. This suggests, under the conditions of that study, that robustness depends on the match between available data dimensionality and model complexity. Whether this principle generalises to clinical 10–20 recordings requires further investigation.

Qin et al. also showed that the effect of sensor loss is spatially non-uniform: removing electrodes near the active generator produces much larger degradation than removing distant electrodes, and deeper sources are more vulnerable. This observation links effective dimensionality to montage geometry and anticipates the targeted-density principle discussed clinically in Section 5.1.

4.6. The unmixing theorem: an example of an algebraically determined solution with $L < M$ sources

A theoretically relevant formulation for low-density EEG arises from work on ROI-based unmixing of distributed inverse solutions (Biscay et al. 2018). The starting point is the general property that, for any linear inverse method, estimated source activity at each location represents a mixture of true sources determined by the resolution matrix, leading to spatial leakage and artificial dependencies in connectivity analyses. (A formal mathematical statement of the theorem and conditions for validity are provided in Supplementary Section S3. That derivation adopts the known/fixed source orientation convention — scalar amplitudes per ROI — under which the theorem permits recovery of up to $L \leq M-1 = 18$ scalar ROI time series with S10-20. For unconstrained 3-D source vectors the recoverable limit is $L \leq (M-1)/3 \approx 6$; see S3b.)

Conceptually, this approach operates within the same low-dimensional regime described in Section 4.5: rather than estimating a high-dimensional distributed solution directly, the problem is projected onto a reduced subspace aligned with the information content of the data. In this sense, it is related to parametric and sparse methods that constrain the number of effective sources, which are more robust to electrode degradation.

An important distinction is that dimensionality reduction can be achieved either through explicit assumptions about the number of active generators (as in dipole fitting or RAP-MUSIC) or through post-hoc projection of distributed estimates onto a predefined regional basis. These strategies are complementary and may be appropriate for different types of analyses, depending on the spatial scale and assumptions required by the scientific question.

4.7. Spatial distribution and information geometry: electrode spacing, Nyquist constraints, and information efficiency

The debate over electrode density has almost exclusively focused on electrode count as if it were the only relevant variable. Yet from the perspective of information theory and sampling theory, the spatial arrangement of electrodes — their geometric distribution across the scalp surface — is at least as important as their total number. Two systems with the same number of electrodes can differ dramatically in the amount of independent information they capture, depending on how uniformly they tile the scalp. The 10–20 system was explicitly designed to achieve near-uniform coverage of the superior scalp surface, and this property has profound consequences for its information geometry.

The physical basis of this argument rests on a well-established property of the head volume conductor: the poorly conducting skull acts as a spatial low-pass filter on cortical currents before they reach the scalp. Srinivasan, Nunez & Silberstein (1998) showed analytically, using a four-concentric-sphere head model and stochastic field theory, that this low-pass filtering introduces substantial artificial coherence between nearby electrodes: pairs of scalp electrodes within approximately 10–12 cm of each other can appear correlated even when the underlying cortical sources are completely independent (Srinivasan et al. 1998). This implies that there is a practical spatial resolution floor: below a critical inter-electrode distance, additional channels

provide increasingly correlated and partially redundant information, though the transition is gradual rather than a sharp threshold.

Nunez and Srinivasan (2006) formalise this constraint in terms of the spatial frequency content of the scalp EEG. Because volume conduction through the skull removes spatial frequencies above approximately 0.1–0.2 cycles per cm, the EEG potential field on the scalp is spatially smooth relative to the cortical surface, with a half-power spatial wavelength of roughly 5–10 cm. By the Nyquist sampling theorem, adequate spatial sampling of this field requires an inter-electrode spacing of approximately half the shortest spatial wavelength of interest — roughly 2.5–5 cm for the dominant spatial frequencies of the clinical qEEG field. The standard 10–20 system achieves an inter-electrode spacing of approximately 6–7 cm (roughly 10–12% of the mean head circumference ≈ 55 cm, varying by axis and landmark). This spacing approaches the sampling requirement for the lower-spatial-frequency portion of the clinically dominant scalp field, but does not fully sample the highest spatial frequencies within the 0.1–0.2 cycles/cm range. Denser arrays therefore retain additional spatial information, although the marginal gain decreases as spatial frequency components approach the noise floor.

The geometric uniformity of the 10–20 system — each electrode at 10–20% intervals along standard scalp lines (Jasper 1958) — and the anatomical consistency of its positions (grand SD 4.6–7.8 mm across subjects; Koessler et al. 2009) provide broad and approximately uniform spatial coverage while limiting local redundancy among neighbouring channels. This uniformity argument applies primarily to the low-spatial-frequency components that dominate clinical qEEG and should not be extrapolated to focal cortical sources or high-frequency gamma-band activity, where fine-grained spatial discrimination is required.

5. Clinical Validation with Low-Density Configurations

The studies reviewed in this section provide clinical evidence organised according to the four spatial-resolution levels of the taxonomy (Section 2.2). Within each subsection, evidence is assessed across signal types (resting-state spectra, evoked potentials, connectivity measures, interictal spikes) and clinical applications. A general caveat applies throughout: most of these studies were not designed as comparative trials pitting 19-electrode against high-density systems in matched samples. Evidence therefore remains largely indirect — demonstrating that low-density pipelines produce clinically useful or neurobiologically plausible results, rather than demonstrating equivalence with high-density alternatives. Direct, controlled comparisons remain limited and are flagged explicitly in Supplementary Tables S6a–S6b and in the individual subsections.

To address the structural conflict of interest — namely that much of the affirmative evidence for Levels 1–3 draws substantially on methods and datasets originating with the author group — Supplementary Table S6a disaggregates the evidence base for each major claim, distinguishing independent from author-group contributions and assigning an explicit evidential status. This transparency is intended to allow readers to form their own judgement about which conclusions are well-founded across multiple independent lines, and which remain provisional.

5.1. Epilepsy [Levels 1–4; resting-state and interictal spikes; clinical observational]

Epilepsy provides the most demanding clinical test because presurgical decisions can require sub-lobar precision. The evidence nevertheless separates two questions clearly: low-density ESI can provide useful lobar-level and decision-support information, but routine focal Level-4 localisation benefits substantially from higher density and individual head models.

Sperli et al. (2006) reported 90% lobar localisation in 30 operated paediatric patients using 19–29 electrodes; the two cases requiring 128 electrodes involved mesial temporal sources, consistent with the inferior-surface coverage limitation. Lantz et al. (2003) and Sohrabpour et al. (2015) showed that the largest localisation gain occurs when density increases from roughly 30 to 64 electrodes, with diminishing returns above 64 (Figure 4; Supplementary Figure S2).

Independent multicentre work extends this picture. Automated ESI with the standard 25-electrode IFCN montage achieved clinically meaningful accuracy and added value in presurgical evaluation (Baroumand et al. 2018; Baroumand et al. 2022; Stirling et al. 2023). These studies establish the utility of low-density ESI as part of a multimodal workflow, not equivalence to high-density focal localisation.

Paediatric and infant studies likewise report useful lobar concordance with fewer than 25 electrodes, while deep-learning approaches may reduce sensitivity to configuration changes (Ricci et al. 2021; Rong et al. 2025; Santalucia et al. 2026).

Systematic reviews place these results in context: lobar concordance is commonly achievable across electrode counts, whereas sub-lobar accuracy depends much more strongly on density, anatomy, head modelling, and the chosen gold standard (Plummer et al. 2008; Michel & Brunet 2019). Accordingly, the 10–20/IFCN range is useful for Levels 1–2 and for clinical decision support, but should not be presented as a standalone substitute for 64–256 channel ESI when the surgical question requires precise Level-4 localisation.

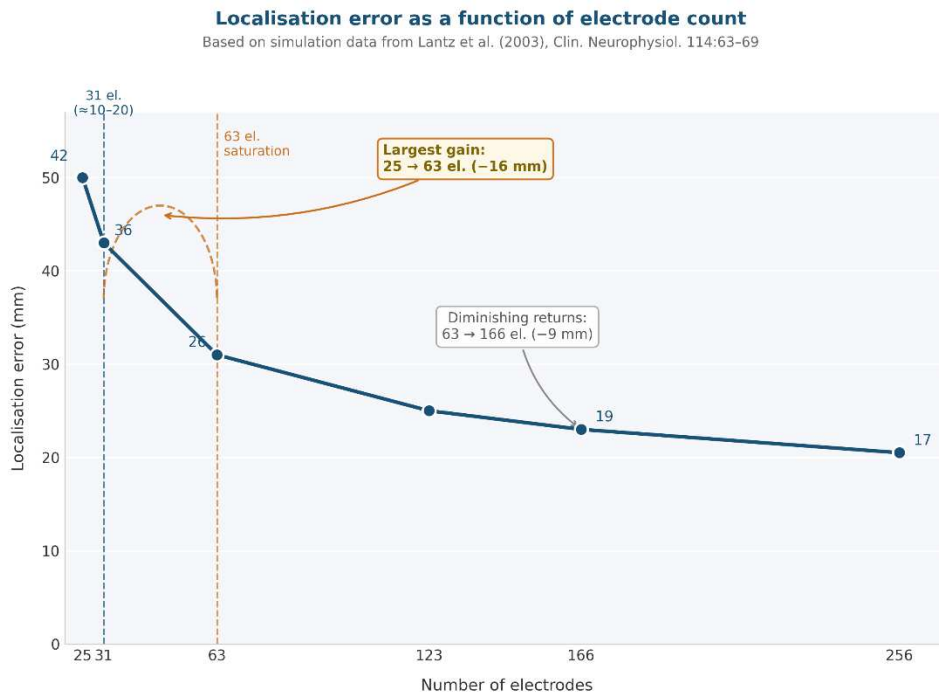


Figure 4: Localisation error as a function of electrode count: the law of diminishing returns. Conceptual schematic illustrating the empirical pattern reported across multiple electrode-density studies. Anchor data points are drawn from Lantz et al. (2003; 14 epilepsy patients, implanted dipole ground truth; 31, 63, and 123 channels) and Sohrabpour et al. (2015; simulated and clinical paediatric epilepsy; 19–256 channels). The extended portion of the curve (above 128 channels) represents a conceptual extrapolation consistent with both sources and is not derived from independent empirical measurements. The curve illustrates the classic law of diminishing returns: the steepest reduction in error occurs between approximately 25 and 64 electrodes, after which incremental gains decrease substantially. This schematic is intended to show the qualitative shape of the relationship, not to provide precise quantitative benchmarks.

The multicentre evidence is important because it shifts the question from idealised localisation error to clinical added value. Baroumand et al. (2018) reported 61% fully automated and 78% semi-automated accuracy with a 25-electrode IFCN montage, comparable in the same cohort to MRI, PET, and SPECT. Stirling et al. (2023) found that automated 25-electrode ESI changed the presurgical decision in 35% of consecutive cases, while Baroumand et al. (2022) demonstrated that ictal low-density ESI adds localising value comparable to conventional imaging modalities. These results do not establish sub-centimetre equivalence to high-density ESI, but they do show that standard-density source imaging can contribute actionable information in a multimodal presurgical pathway.

Targeted density provides a clinically attractive compromise when the suspected region is known. Horrillo-Maysonnial et al. (2023) obtained 93% sublobar concordance with high-density EEG after adding 8–11 strategically placed electrodes to a 25-electrode base montage (median distance 13.2 mm). Earlier work with sphenoidal and insular coverage supports the same anatomical principle: local sampling can matter more than a uniform global increase when the missing information arises from a specific surface gap. This is consistent with the failure patterns reported for mesial temporal and interhemispheric sources.

Recent paediatric and infant studies broaden the evidence base. Low-density ESI has shown clinically meaningful localization accuracy in surgical epilepsy cohorts. Vorderwülbecke et al. (2021) reported comparable diagnostic accuracy for interictal source localization across 25–

257 electrode configurations, while Santalucia et al. (2026) found accuracies of 66% for interictal ESI, 72% for ictal ESI, and 75% for their combination in children undergoing surgery before 7 years of age. In a separate paediatric cohort, Ricci et al. (2021) showed that low-density ictal ESI approximated the intracranially defined seizure-onset zone and that its concordance with resection predicted postsurgical seizure freedom. Deep-learning approaches appear more robust to configuration variation but still show degradation below practical density thresholds (Rong et al. 2025; Abbas et al. 2023). Detailed study-level results and evidential status are provided in Supplementary Section S6.

5.2. Dementia and cognitive impairment [Levels 1–2; resting-state spectra and source power; clinical observational]

Quantitative EEG with S10-20 has a substantial evidence base in Alzheimer’s disease (AD) and mild cognitive impairment (MCI). LORETA studies have linked lobar alpha-source changes to hippocampal atrophy, cognition, and disease severity (Babiloni et al. 2009; Smailovic & Jelic 2019), while multicentre cohorts have repeatedly identified parieto-occipital delta/alpha abnormalities and source-level connectivity changes using 19–21 electrodes (Lizio et al. 2011; Vecchio et al. 2014).

The strongest interpretation of this literature is at Levels 1–2. A systematic review of ML-based EEG for Alzheimer’s diagnosis reported that one included 19-channel study achieved 90.9% sensitivity and 76.7% specificity against amyloid PET (Park et al. 2023, reviewing multiple primary studies); this primarily validates low-density spectral biomarkers rather than anatomical precision. Similarly, source-projected measures can show biologically plausible relations to MRI and cognition without implying sub-lobar accuracy.

Across independent cohorts, diagnostic and correlational performance supports adequacy of standard-density EEG for spectral biomarkers and lobar source-power questions. What remains largely absent is a matched head-to-head test showing whether high-density source imaging improves sensitivity or specificity for the same dementia biomarkers in the same participants. Thus, the evidence supports Level 1–2 use but not focal anatomical claims.

Several independent cohorts support source-projected Level 2 measures. Lizio et al. (2011) used 19-channel LORETA in a multicentre cohort of 378 participants and reported delta/alpha source-power differences across healthy ageing, MCI, and AD. Vecchio et al. (2014) found alpha-frequency frontoparietal source-connectivity differences between AD and MCI, and van der Hiele et al. (2007) reported successful lateralisation of hemispheric asymmetries with 21 electrodes. These studies are valuable because they demonstrate reproducible regional source patterns outside the present authors' methodological lineage.

At the same time, dementia literature illustrates an important boundary of inference. A clinically useful low-density biomarker may be entirely valid even if its source map is spatially blurred. Diagnostic accuracy, correlation with structural MRI, and longitudinal reproducibility establish biomarker validity at the relevant spatial scale; they do not demonstrate that a 19-electrode inverse map has the same anatomical precision as a high-density reconstruction.

5.3. Test-retest reliability of qEEG source estimates [Levels 1–2; resting-state spectra; psychometric]

Reliability is a necessary condition for clinical use. Cannon et al. (2012) reported 30-day ICCs of 0.81–0.82 for 19-channel LORETA measures across eight ROIs. Later work confirms that reliability is frequency-dependent: alpha and beta source power are generally the most reproducible, whereas delta, gamma, and aperiodic measures require greater caution (Ding et al. 2022; Popov et al. 2023). Source-space estimates can equal or exceed sensor-level reproducibility, supporting longitudinal Level 1–2 applications without implying high spatial precision.

Frequency specificity is therefore central when low-density source measures are proposed for monitoring. Popov et al. (2023) reported excellent reliability for alpha and beta measures, but much lower values for delta and gamma, and Ding et al. (2022) similarly found that reproducibility varies across spectral components and recording conditions. The appropriate claim is thus not that all 19-channel source estimates are highly reliable, but that selected spectral features—particularly alpha-band regional power—can achieve psychometric stability suitable for longitudinal Level 1–2 applications.

5.4. Effective connectivity in paediatric neurodevelopmental disorders [Level 3; connectivity; clinical observational]

Bosch-Bayard et al. (2020) analysed directed connectivity in 227 children with developmental reading disorders using S10-20, sLORETA, and isolated effective coherence among 17 ROIs. Group-specific hubs were neurobiologically interpretable and satisfied the $L < M$ dimensionality constraint of the unmixing framework. The result is consistent with Level-3 feasibility, but it is not a ground-truth validation, and no matched high-density comparison has been published.

Independent work provides convergent, though methodologically different, evidence. RAP-MUSIC modelling of the P50 ERP identified posterior-cingulate differences in dyslexia using a standard montage (Stoitsis et al. 2008b), and the broader dyslexia literature repeatedly reports altered long-range synchrony in left-hemisphere language networks (Vandermosten et al. 2012). These findings support the neurobiological plausibility of sparse-array network results, while leaving their spatial accuracy relative to high-density EEG unresolved.

In Bosch-Bayard et al (2020), isolated effective coherence was calculated among 17 ROIs selected within the available dimensionality of the 19-electrode recording, and group differences were assessed across the frequency spectrum. The left calcarine sulcus and left rolandic operculum emerged as group-specific hubs. Because the clinical comparison did not include an independent anatomical ground truth, these findings should be interpreted as proof of neurobiological and statistical coherence rather than direct validation of sub-lobar localisation.

The independent P50 study of Stoitsis et al. (2008b) is useful precisely because it used a different inverse strategy—RAP-MUSIC dipole modelling—and nevertheless identified a plausible posterior-cingulate generator discriminating dyslexic children from controls. Convergence across inverse families reduces concern that the observed group differences are

peculiar to a single pipeline, although it still does not answer how much spatial information would be gained in the same participants with high-density EEG.

5.5. Normative connectome development during the first year of life [Level 3; connectivity; normative observational]

Bosch-Bayard et al. (2022b) tested whether a 19-electrode, ROI-constrained pipeline could track normal connectivity maturation across the first year of life. In 125 healthy infants, sLORETA estimates from 16 ROIs were unmixed and analysed with isolated effective coherence.

The resulting developmental sequence—posterior-to-anterior and subcortical-to-cortical progression, with prominent thalamocortical changes—was consistent with established myelination trajectories. This constitutes neurobiological plausibility rather than anatomical ground truth. Independent simulations and developmental connectivity studies likewise indicate that predefined, well-separated ROI networks are more tolerant of density reduction than blind whole-brain decomposition, provided leakage correction and accurate forward modelling are used (Boersma et al. 2011; Allouch et al. 2023; Hatlestad-Hall et al. 2023).

The infant study used 16 ROIs, satisfying the $L < M$ condition, and assessed age-dependent changes across delta, theta, alpha, beta-1, and beta-2 bands. The largest developmental changes occurred in theta and alpha, with posterior-to-anterior progression and thalamic hub behaviour consistent with established thalamocortical maturation. The strength of this evidence lies in concordance with an independently known developmental sequence rather than in voxel-level anatomical validation.

Independent developmental and simulation studies set useful bounds. Boersma et al. (2011) reported age-related source-connectivity changes consistent with maturation using a different pipeline. Allouch et al. (2023) showed that inverse-method and connectivity-measure choice can partly compensate for density reductions, and Hatlestad-Hall et al. (2023) found that predefined ROI graph measures tolerate reduction from 128 to 64 channels better than unconstrained whole-brain network identification. Together, these results support targeted Level-3 analysis while reinforcing the need for explicit parcellation, leakage correction, and conditioning checks.

5.6. Event-related potentials and steady-state responses [Levels 1–4; evoked potentials; theoretical and observational]

Evoked responses differ from resting-state EEG because trial averaging can markedly improve SNR. The same spatial taxonomy nevertheless applies: broad components can support Level 1–2 inference with sparse, uniformly distributed montages, whereas focal generators remain density-limited.

Frequency-domain VARETA studies of steady-state responses localized expected occipital/parieto-occipital or auditory generators with 30–47 electrodes (Müller et al. 1998; Picton et al. 2003). Michel et al. (2004) showed why geometry matters: a non-uniform 19-electrode occipital subset mislocalised a visual source, whereas 19 uniformly distributed electrodes recovered the correct occipital lobe. This directly supports the distinction between electrode number and spatial coverage.

For transient ERPs, 10–20 system LORETA-family studies have produced neuroanatomically plausible N1 and P300 source patterns (Anderer et al. 1998; Mulert et al. 2004). High-density studies provide convergent evidence from a richer spatial basis: Tsolaki et al. (2017) localised MMN and P300 generators in MCI and Alzheimer’s disease using an explicit high-density EEG approach, confirming the neuroanatomical plausibility of the sources but not establishing sufficiency of the 10–20 system for this task. High-density subsampling studies further show that performance depends jointly on method, SNR, source multiplicity, and electrode density (Yao & Dewald 2005). Broad late components therefore map naturally to Levels 1–2; focal early components such as N170 or restricted MMN generators fall toward Levels 3–4.

ERP-specific advantages should not be overgeneralised: averaging changes the noise regime, temporal sequencing can help separate generators, and direct matched-density comparisons remain sparse. The evidence supports lobar attribution from 19 uniformly distributed channels for broad components, not routine sub-lobar localisation.

Steady-state responses provide a particularly favourable test because signal energy is concentrated at the stimulation frequency and its harmonics. Müller et al. (1998) used frequency-domain VARETA with a 30-electrode configuration containing standard 10–20 positions and recovered occipital/parieto-occipital sources and attention-related extrastriate modulation consistent with independent physiology. Picton et al. (2003) similarly localised auditory steady-state generators with 47 electrodes. These examples support regional source attribution near the Level 2 scale rather than focal equivalence with high-density ESI.

ERP inference also benefits from temporal structure: sequentially active generators may be separable by latency even when their scalp fields overlap spatially. This advantage does not remove the spatial sampling limit. Direct matched comparisons across 19, 32, 64, and high-density configurations remain scarce, especially for focal early components. A prospective density study spanning broad P300/N1-type components and focal N170/MMN-type generators would provide a direct test of the taxonomy.

5.7. Practical decision guide: choosing electrode density by spatial level and signal type

The taxonomy and evidence support an application-specific decision rule rather than a universal channel threshold.

Use S10-20 when the target is sensor-level spectral biomarkers or lobar source power (Levels 1–2), several-centimetre regional spatial spread is acceptable, data quality and batch effects are controlled, and inferior-surface coverage is not central to the question.

For Level-3 ROI connectivity, use S10-20 only with an explicitly low-dimensional model ($L < M$), anatomically separated ROIs, verified conditioning, leakage control, and an accurate forward model. Under the scalar/fixed-orientation convention this allows up to $L \leq 18$ ROIs; for free 3-D source vectors the practical limit is $L \leq 6$. Results should remain provisional pending direct validation.

Use ≥ 64 electrodes with subject-specific head modelling for routine focal presurgical localisation, blind spatial decomposition of distributed networks, or other questions requiring

sub-lobar precision. Targeted inferior or zone-specific electrodes may be a more efficient intermediate solution when the limitation is local coverage rather than global density.

Supplementary Section S6 provides claim-level and study-level evidence tables so that these recommendations can be traced to their empirical basis.

5.8. Subcortical source estimation with the 10–20 system: feasibility and limitations

Subcortical source estimation is biophysically possible only for generators whose geometry, amplitude, and synchrony produce a measurable scalp field. Depth alone is not decisive: open-field structures such as parts of the hippocampus are more visible than closed-field generators, while model accuracy becomes increasingly important (Attal & Schwartz 2013; Piastra et al. 2021).

For the 10–20 system, evidence is limited. Most direct demonstrations use high-density EEG or MEG, and a cortical-only source model will necessarily project subcortical activity to the cortex. Low-density subcortical claims therefore require high SNR, an explicit subcortical forward model, and cautious interpretation; detailed evidence and caveats are provided in Supplementary Section S7.3.

6. Five Decades of Normative Research: The Most Extensive Natural Experiment

Five decades of normative qEEG provide a long-term stress test of what the 10–20 system can support at Levels 1–2. Developmental neurometrics, the Cuban Human Brain Mapping Project, and later multinational harmonisation studies have produced reproducible age- and disease-related spectral patterns from standard montages (John et al. 1977, 1980; Bosch-Bayard et al. 2001, 2020a, 2020b, 2026; Li et al. 2022).

The strongest inference from this history is not that S10-20 provide high-resolution tomography, but that they can support stable regional and spectral inference across populations when acquisition, reference, forward modelling, and normative statistics are controlled. Independent replication of some author-group pipelines remains limited and is treated explicitly in Supplementary Section S6.

More recent connectome-informed approaches such as ξ - α NET suggest that strong structural and biophysical priors may further narrow effective source uncertainty from 19-electrode cross-spectra. Because current evidence comes from the developing author group, these results should be regarded as hypothesis-generating until independently replicated; the detailed benchmark and Supplementary Figure S3 are provided in Supplementary Section S7.2.

HarMNqEEG extends this natural experiment to 1,564 participants from 9 countries, multiple devices, and 14 studies using compatible standard montages (Li et al. 2022), and its methodological implications for harmonisation are taken up in Section 7.

7. Batch Effects and Technical Harmonisation in Multinational qEEG

Multisite qEEG introduces technical variation from amplifiers, preprocessing, and study-specific protocols. HarMNqEEG showed that explicit harmonisation of these batch effects materially improves normative comparability and diagnostic performance (Li et al. 2022). This does not make electrode density irrelevant; rather, it shows that adding channels cannot compensate for unmodelled site effects.

Within this framework, the methodological choices for a researcher or clinician using the 10–20 system depend on the inference goal. For normative qEEG and spectral biomarkers (Levels 1–2): (1) batch-effect correction and spectral normalisation (Global Scale Factor, Hernández et al. 1994; ComBat variants; HarMNqEEG norms) and (2) selection of inverse methods with guaranteed localisation properties (eLORETA, sLORETA; Biscay et al. 2018 unmixing) are the highest-leverage interventions; atlas head models (MNI-space) reduce co-registration error when individual MRI is unavailable. Increasing electrode density addresses different error sources — spatial resolution and CTF breadth — and its priority depends on the spatial precision required by the question: it is less critical for lobar-scale normative analyses but clearly indicated for sub-lobar or focal applications (Levels 3–4). Increasing density without correcting batch effects can produce less reproducible results than properly harmonised 10–20 qEEG, because harmonisation and density address separate sources of variance.

The Global Scale Factor described by Hernández et al. (1994) illustrates the magnitude of such technical variance: a common multiplicative component accounted for a large fraction of age-corrected log-spectral variance across derivations and frequencies. Removing this global factor improves comparability without adding electrodes. The broader lesson is that spatial sampling and technical harmonisation solve different problems and should not be traded against each other.

8. When the 10–20 System Is Genuinely Insufficient

A balanced review requires precisely delineating the contexts in which the 10–20 system effectively falls short. These contexts can be delineated according to the spatial requirements of the application, and in most cases have specific solutions that do not require replacing the entire system.

8.1. Inferior surface coverage: a major geometric limitation

The most consistently documented limitation is coverage of the inferior brain surface — specifically the basal temporal cortex and orbitofrontal region. As Song et al. (2015) demonstrated, adding sensors on the inferior surface improves source localisation accuracy for deep sources across all electrode densities, including sparse arrays. Sperli et al. (2006) found that 128 electrodes were necessary in only 2 of 30 patients, both with mesial temporal lobe epilepsy. This illustrates that inferior coverage is an important, specifically remediable weakness of the 10–20 montage; Song et al. also found, however, that greater sensor density improves localisation globally, with maximum accuracy at whole-head dense sampling, so

adding inferior electrodes and increasing density are complementary, not competing, strategies. The IFCN 25-electrode extension addresses this gap with six inferior temporal positions (F9/F10, T9/T10, and P9/P10) (Figure 5). Because these electrodes lie closer to craniofacial and neck musculature, their use requires explicit inspection for electromyographic contamination and the same channel-level artefact control applied elsewhere in the recording; the added anatomical coverage does not remove the need for signal-quality assessment (Seeck et al. 2017).

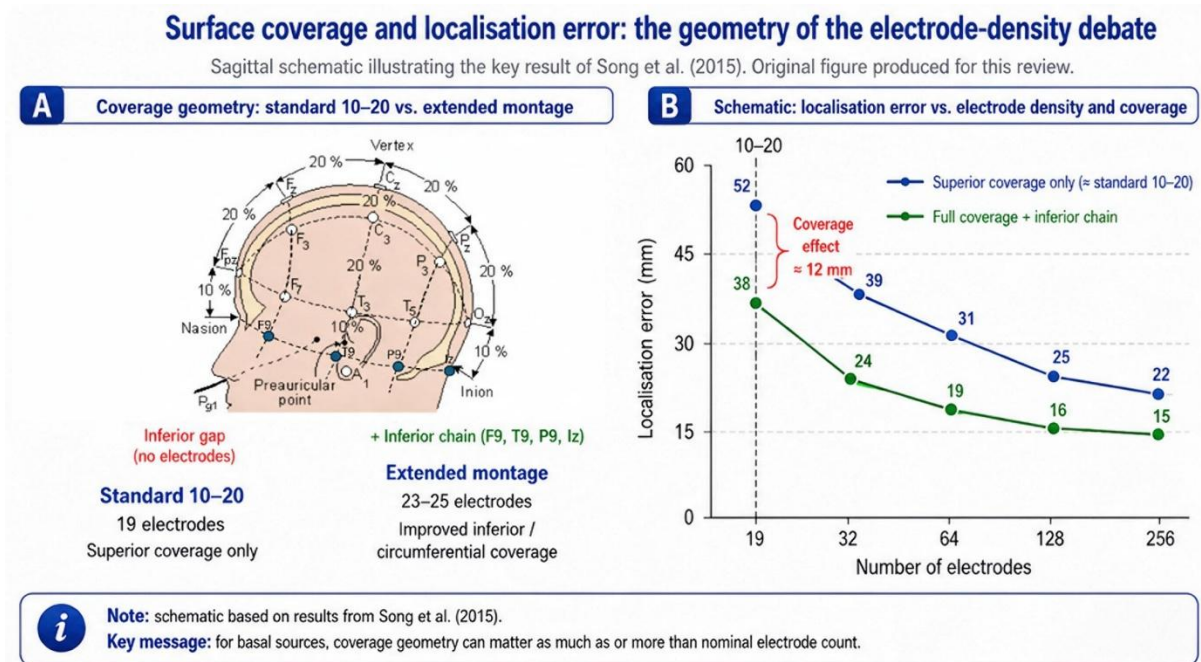


Figure 5: Surface coverage geometry and localisation error: the standard 10–20 montage and the inferior surface gap.

8.2. Large-scale resting-state network detection by sICA: high density matters

Liu et al. (2018) examined the detection of resting-state networks (RSNs) from EEG using spatial independent component analysis (sICA) as a function of electrode density, head model accuracy, and source localisation algorithm. Starting from a 256-channel reference, they subsampled recordings to 128, 64, and 32 channels. Their repeated-measures ANOVA revealed that electrode density has the strongest effect on RSN reconstruction quality ($F = 29.26$, $p < 0.0001$), followed by head model accuracy ($F = 11.12$, $p = 0.0029$) and source localisation method ($F = 7.99$, $p = 0.0085$). With 32 channels, most EEG-RSN maps showed substantial degradation in spatial correspondence with fMRI-RSN templates.

Three important qualifications contextualise this finding for the present argument. First, Liu et al. (2018) used sICA over continuous power envelopes as their detection method — an approach that requires the full spatial diversity of source signals across many ROIs simultaneously to achieve blind decomposition. This is a fundamentally more demanding task than targeted analysis of a predefined set of ≤ 18 ROIs, which is what the unmixing theorem of Biscay et al. (2018) addresses. Second, Liu et al. found that eLORETA and sLORETA were the best-performing source localisation algorithms across all electrode densities, substantially outperforming MNE and LCMV — confirming our argument that method choice is as

important as electrode density. Third, RSNs with less distributed spatial patterns (such as the dorsal somatomotor network) could be successfully reconstructed even with low-density montages, while deeper or more distributed networks (such as the DMN) were more affected by electrode reduction. This network-specificity argument maps directly onto the qEEG application domain: lobar spectral indices and targeted effective connectivity between predefined regions (as in Bosch-Bayard et al. 2020) are less affected by electrode density than blind whole-brain network decomposition. Figure 6 summarizes the three aspects discussed in this section.

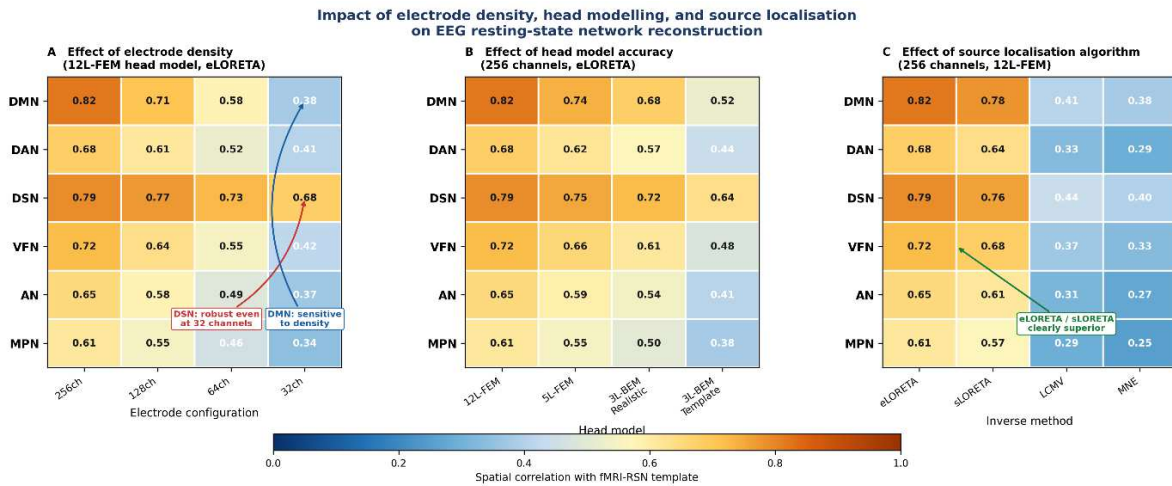


Figure 6: Impact of electrode montage density, head modelling, and source localisation on EEG resting-state network reconstruction. Panel A: effect of electrode density (256→128→64→32 channels; 12L-FEM, eLORETA). Panel B: effect of head model accuracy (12L-FEM, 5L-FEM, 3L-BEM realistic, 3L-BEM template; 256 ch, eLORETA). Panel C: effect of source localisation algorithm (eLORETA, sLORETA, LCMV, MNE; 256 ch, 12L-FEM). Colour scale: spatial correlation with fMRI-RSN template (0 = no correspondence; 1 = perfect correspondence). DMN = default mode; DAN = dorsal attention; DSN = dorsal somatomotor; VFN = visual foveal; AN = auditory; MPN = medial prefrontal. FEM = finite element method; BEM = boundary element method.

Complementarily, Hassan and Wendling (2018) provide a tutorial overview of EEG source connectivity, noting that many studies have found that at least 128 electrodes are needed for satisfactory network identification when minimum-norm class methods are used. This recommendation, however, is explicitly framed in the context of dense-array source connectivity for network science applications — not for the targeted qEEG analyses, clinical biomarker extraction, or ROI-based effective connectivity that characterise the bulk of normative qEEG research. The minimum electrode requirement is application-dependent, not universal.

8.3. Presurgical epilepsy planning: high-density, clearly justified

For sub-lobar presurgical localisation (precision < 2 cm), particularly in epilepsy surgical planning, the evidence clearly supports high-density recording. Brodbeck et al. (2011) documented in 152 operated patients that source imaging sensitivity and specificity drop from 84%/88% with 128–256 channels and individual MRI to 57%/59% with sparse arrays below 32 channels and a template head model. This specific context warrants high-density recording and represents the appropriate use case for 64–256 channel systems.

9. The 10–20 System as a Tool for Equitable Global Neuroscience

Accessibility matters because standard EEG remains the only broadly available functional neuroimaging modality in many resource-limited settings. The taxonomy provides a principled way to preserve access without lowering scientific standards: use low-density systems where Levels 1–2 inference has been demonstrated fit-for-purpose, add targeted electrodes when geometry is the limiting factor, and require high density when the clinical or scientific question genuinely demands Level-4 precision (McLane et al. 2015).

Equity is therefore an argument for task-appropriate technology, not for treating S10-20 as universally adequate.

This argument has a strict boundary. Limited resources cannot convert a Level-4 question into a Level-2 question: presurgical focal localisation, fine-grained network decomposition, and other spatially demanding tasks still require the density and modelling supported by the evidence. The role of the 10–20 system is to maximise valid access where the requested inference genuinely lies within its supported scale.

10. Limitations of this review

Several limitations should be acknowledged. First, although the literature search was structured and PRISMA-informed, this is a critical narrative review rather than a pre-registered systematic review or meta-analysis; study selection and synthesis therefore remain partly interpretive. Second, several methods discussed favourably—including sLORETA/eLORETA, VARETA, unmixing, HarMNqEEG, and ξ - α NET—were developed by or in close collaboration with current authors. We mitigate this by separating independent from author-group evidence, bounding conclusions by validity level, and providing claim- and study-level evidence tables in Supplementary Section S6.

Third, the clinical literature is heterogeneous in design, sample size, head model, and gold standard, and matched low- versus high-density comparisons in the same cohorts are uncommon. Fourth, evidence is strongest for resting-state spectral/qEEG and epilepsy; ERP, connectivity, and subcortical applications are less uniformly validated. Fifth, improvements in electrodes, amplifiers, head modelling, and artefact control may shift the practical noise ceiling over time. These limitations make prospective, matched-density comparisons across all four validity levels the most important next step.

11. Conclusions

This review has examined the conditions under which the 19-channel 10–20 electrode system can support valid EEG source estimation. The conclusions are application-specific and organised according to the four-level validity taxonomy introduced in Section 2.

For sensor-level spectral biomarkers (Level 1) and lobar source-power inference (Level 2), the available evidence demonstrates utility of 19-electrode systems under appropriate conditions: inverse methods with zero PSF-peak localisation bias under the assumed forward model (eLORETA, sLORETA), adequate noise control, and explicit acknowledgment that spatial

resolution is limited to the lobar scale (SD 20–50 mm (eLORETA/sLORETA; note that the lower bound ~20 mm derives from a macaque ECoG/EEG model, see Table 2 footnote [c]) and FWHM 30–35 mm (cLORETA/VARETA; volumetric metric, see Table 2 footnote [d])). This does not imply equivalence to high-density EEG, nor that density increases would not further improve performance; studies comparing eLORETA at 18 versus 64 versus 173 channels find monotonic improvement (Dattola et al. 2020), and matched head-to-head comparisons in clinical cohorts remain largely unavailable. Specifically for Level 2, the DLE = 0 property of eLORETA/sLORETA ensures the PSF peak is correctly positioned but does not quantify CTF-mediated contamination of lobar power estimates from adjacent generators — a gap that direct 19-vs-64 lobar-power comparisons would close. The evidence of utility reviewed here is therefore convergent rather than definitive. [Empirical basis: normative and clinical literature reviewed in Sections 5.1–5.3 and 5.5; theoretical basis: resolution matrix analysis in Sections 4.1–4.3.] Both conclusions rest on indirect evidence — agreement between normative patterns and known neurophysiology, and clinical diagnostic accuracy — not direct matched-density comparison.

For ROI-to-ROI connectivity (Level 3), S10-20 may be feasible when the number of regions does not exceed $M-1 = 18$ scalar ROI time series (assuming known/fixed source orientation; for unconstrained 3-D vectors the bound is $(M-1)/3 \approx 6$), and the lead field column independence condition is satisfied. [Theoretical basis: algebraic unmixing theorem, Section 4.6 and Supplementary S3; the theorem is a proposition of the group and has not been independently verified by external replication in the connectivity context.] Existing clinical findings (Sections 5.4–5.5) are consistent with this framework but constitute convergent rather than direct validation: they demonstrate neurophysiologically plausible patterns rather than accuracy against an independent ground truth. At present, Level 3 applications with S10-20 are best regarded as proof-of-concept pending external validation.

For presurgical focal source localisation (Level 4), the available evidence is consistent across multiple independent studies (Section 5.1): S10-20 are insufficient for reliable sub-lobar localisation, and high-density recordings with subject-specific head models are indicated. This is the conclusion most firmly supported by direct empirical comparison.

A key geometric limitation of the standard 10–20 system is incomplete inferior temporal coverage, affecting localisation of mesial temporal and basal frontal sources. This limitation can be partially mitigated by adding inferior temporal electrodes, as recommended by the IFCN.

Importantly, zero localisation bias in eLORETA/sLORETA does not imply high spatial precision. Source estimates remain spatially extended, and cross-talk between sources persists. For Level 2 applications in particular, the zero-DLE property is a statement about PSF peak positioning, not about the accuracy of estimated lobar source power: the cross-talk function (CTF) at S10-20 is broader than at higher densities, so power attributed to a given lobe absorbs contributions from neighbouring generators. The magnitude of this CTF-mediated contamination under typical clinical noise has not been directly quantified for the 19-electrode case, and constitutes the principal unresolved evidential gap for the Level 2 claim. These limitations must be explicitly acknowledged in sparse-array source imaging.

Overall, the question is not ‘are 19 electrodes sufficient?’ but ‘for which inference tasks is there evidence of utility, and how does performance scale with density?’ The direct density comparisons available — Ryyänen et al., Nguyen-Danse et al., Song et al., Dattola et al. — consistently show real gains with higher density; what the literature does not yet provide is matched head-to-head comparisons across all four validity levels in clinical cohorts. This review synthesises the available evidence of utility without claiming equivalence. The conclusions at Level 1 (sensor-level biomarkers) and Level 4 (focal presurgical localisation) rest on the most substantial empirical foundations, in opposite directions. Those at Levels 2 and 3 are better supported by theoretical argument and convergent clinical evidence than by direct head-to-head comparison and should be interpreted accordingly. The field would benefit substantially from prospective studies comparing 19-electrode and high-density pipelines on matched cohorts across all four validity levels.

Glossary of abbreviations and technical terms

A complete glossary of abbreviations and technical terms is provided in Supplementary Section SG.

Statements and Declarations

Author contributions

Jorge Bosch-Bayard: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing; Rolando Jose Biscay: Methodology, Conceptualization, Writing – original draft; Ronaldo Garcia-Reyes: Methodology, Conceptualization, Writing – original draft; Lidice Galan-Garcia: Methodology, Conceptualization, Writing – original draft; Qing Wang: Conceptualization; Writing – review & editing; Jochem Rieger: Conceptualization; Writing – review & editing; Faisal Mushtaq: Conceptualization; Writing – review & editing; Pedro A. Valdes-Sosa: Conceptualization, Methodology, Formal analysis, Writing – review & editing.

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Conflict of interest

Several methods discussed in this review — including sLORETA, eLORETA, the unmixing procedure, and VARETA — were developed by researchers with whom some of the authors have professional or institutional relationships. Specifically, the unmixing theorem of Biscay,

Bosch-Bayard & Pascual-Marqui (2018) involves a co-author of this review. The clinical evidence reviewed in Sections 5.4 and 5.5 was produced by groups with which the authors have collaborative ties. These relationships constitute a potential bias toward positive findings from the methods and pipelines described, and readers should weigh the evidence accordingly. The taxonomic framework and the critical assessment of conditions and limitations were designed explicitly to mitigate this risk, and external critical references (Grave de Peralta Menendez & Gonzalez Andino 1998; Grova et al. 2006; Plummer et al. 2008; Hatlestad-Hall et al. 2023) have been included to provide independent perspectives. No financial conflicts of interest exist. The authors have no shareholding or consultancy relationships with EEG manufacturers or commercial entities related to the content of this review.

Ethics approval

This is a review article. No primary data collection was performed. No ethics approval was required.

Data availability

No new data were generated or analysed in support of this review.

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